

OpenPowerlifting Dataset Analysis Report

Global Competition Data — 1974 to 2018

8,482 Meets · 386,414 Athletes · 45 Countries · 60 Federations

9 Analyses · Python + pandas + matplotlib

OVERVIEW

Executive Summary

This report presents a full data analysis of the OpenPowerlifting public dataset, covering sanctioned powerlifting competitions held worldwide between 1974 and 2018. The dataset captures individual athlete results across three lifts — Squat, Bench Press, and Deadlift — along with meet metadata including federation, country, date, and equipment category.

Nine analyses were completed, covering gender performance, equipment impact, disqualification patterns, sport growth over time, federation rankings, country-level participation, age group performance, the historic shift from equipped to raw powerlifting, and a numeric correlation heatmap. Key findings are summarised below.

#	Analysis	Key Finding
1	Gender Comparison	Males lift 69% more in Total but only 6.5% more on Wilks
2	Equipment Breakdown	Wraps athletes outscore Multi-ply on Wilks despite lower Total
3	DQ Analysis	22,984 DQ entries; XPC leads at 18.8%; 3,548 repeat offenders
4	Sport Growth	Participation grew 10× from 2011 to 2017 (~8K to 101K/year)
5	Federation Rankings	IPF averages 578 kg Total and 437 Wilks — far above domestic feds
6	Country Breakdown	USA dominates with 256K athletes; Norway 3rd despite 5M population
7	Age Groups	Junior (18-23) and Open (24-39) nearly identical; decline at 40-49
8	Raw Revolution	Raw grew from 1% (1998) to 66% (2014) — complete transformation
9	Correlation Heatmap	Wilks/BodyweightKg correlation = 0.21, confirming normalisation works

Gender Score Comparison

Valid competition entries — excluding disqualifications, no-shows, and missing totals — were grouped by gender and compared across average Total (kg) and average Wilks score. The Wilks metric normalizes strength for bodyweight, making it the fairest tool for cross-gender comparison.

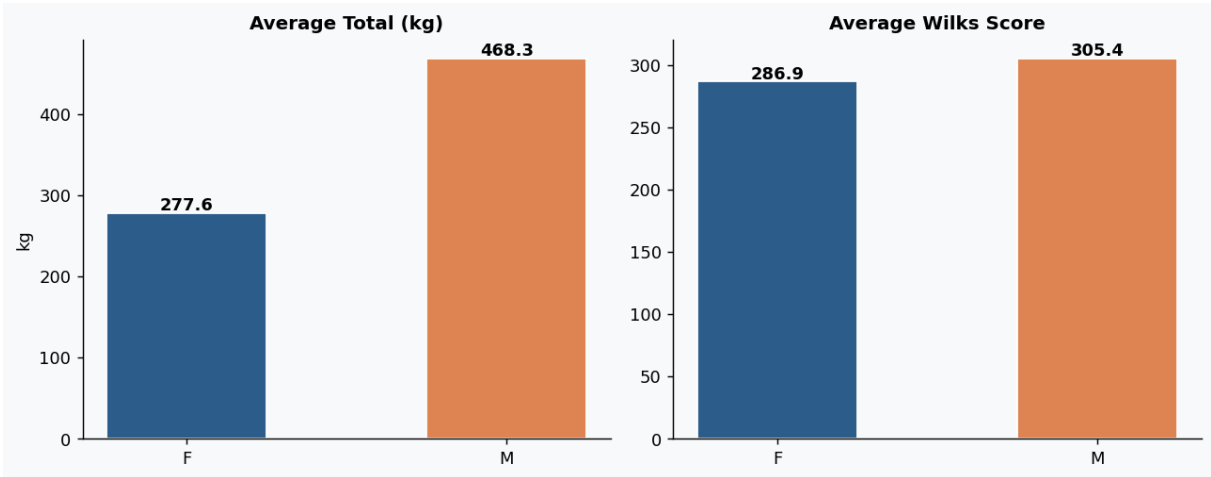


Figure 1 — Average Total (kg) and Average Wilks Score by Gender. Left: raw totals. Right: bodyweight-adjusted Wilks scores.

Gender	Participants	Avg Total (kg)	Avg Wilks	Avg Bodyweight (kg)
Female	84,449	277.6	286.9	69.4
Male	278,785	468.3	305.4	92.0

Key Finding: Males lift 69% more in raw Total on average, but the Wilks gap shrinks to just 6.5% (305 vs 287). Wilks effectively accounts for the bodyweight difference between genders. Male entries outnumber female entries 3.3 to 1, reflecting the sport's historical gender composition — though female participation has grown significantly in recent years.

Equipment Breakdown

Powerlifting sanctions five equipment categories. Supportive gear — single-ply or multi-ply suits and knee wraps — adds significant kilograms to every lift by storing and returning energy during the movement. This analysis compares participant counts and average performance across all equipment types.

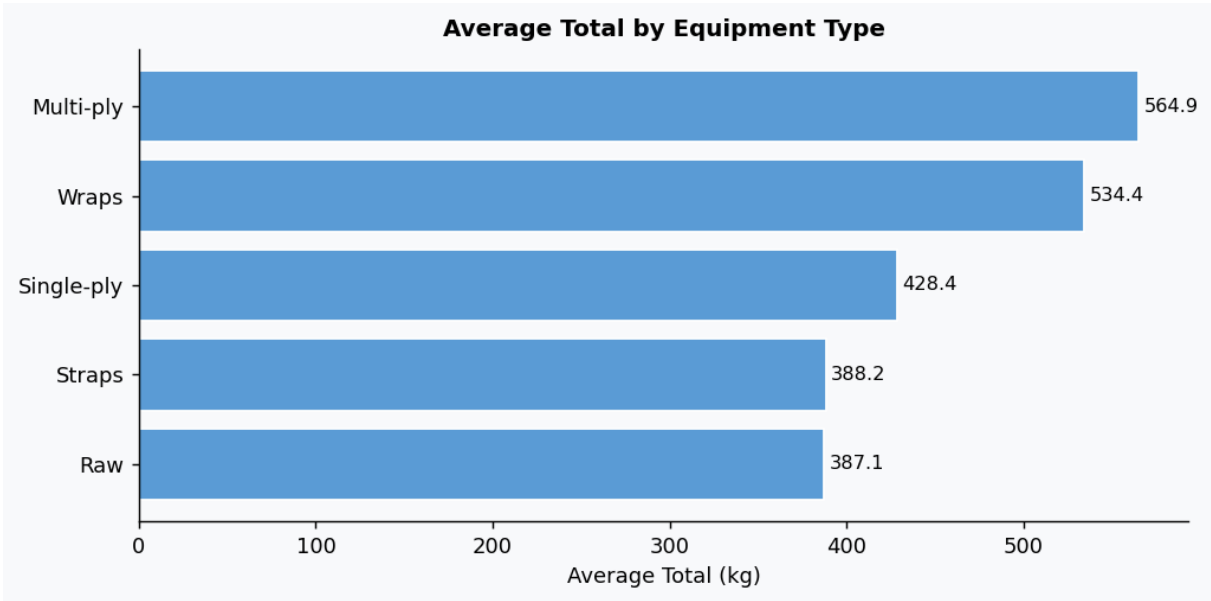


Figure 2 — Average Total (kg) by Equipment Type, sorted ascending. Raw is the baseline; Multi-ply produces the highest totals.

Equipment	Count	Avg Total (kg)	Avg Wilks
Multi-ply	8,416	564.9	362.6
Wraps	44,280	534.4	365.6
Single-ply	130,619	428.4	300.1
Raw	179,912	387.1	283.1

Key Finding: Wraps athletes score higher on Wilks (365.6) than Multi-ply athletes (362.6) despite lifting less in absolute Total — suggesting Wraps competitors are stronger relative to bodyweight. Raw powerlifting is now the most common category with 46% of all valid entries.

Note: Never compare Totals directly across equipment categories. Supportive gear adds 50–150+ kg to a lifter's Total. Wilks is the only fair metric for cross-equipment comparisons.

Disqualified Athlete Analysis

A disqualification (DQ) occurs when an athlete fails all three attempts in a lift, violates equipment rules, or commits a technical infraction. This analysis examines total DQ volume, repeat offenders, and which federations produce the highest disqualification rates.

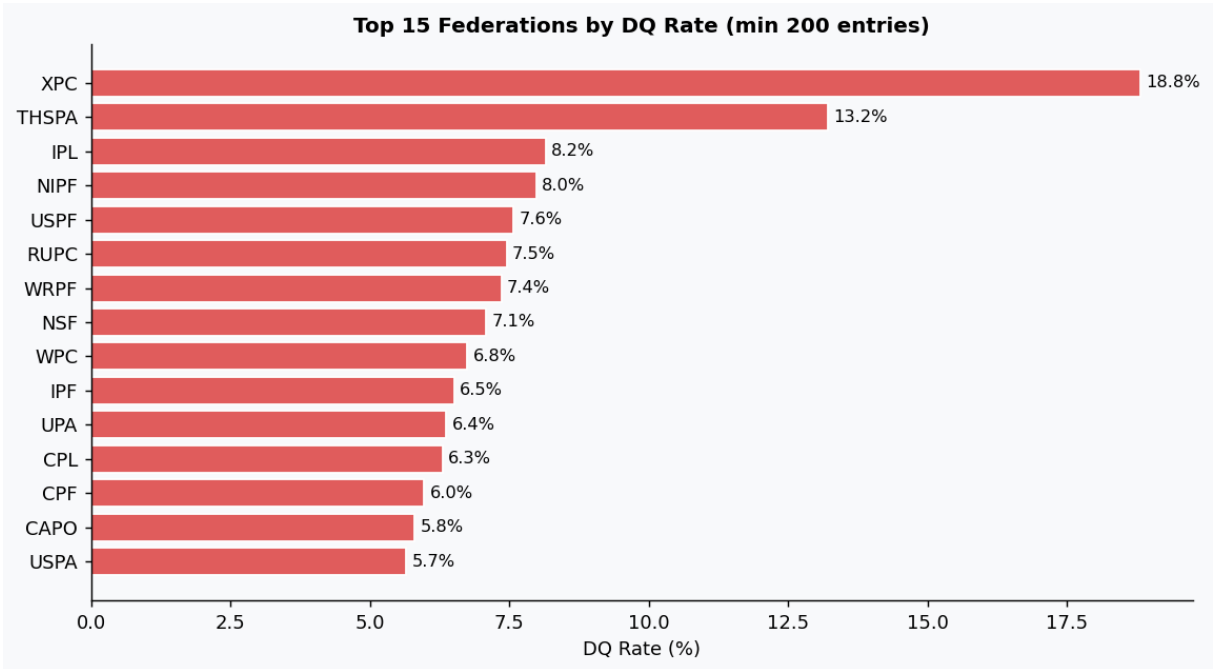


Figure 3 — Top 15 Federations by DQ Rate (minimum 200 entries). XPC leads at 18.8%; THSPA at 13.2% across 60,000+ entries.

Metric	Value
Total DQ entries	22,984
Unique athletes DQ'd	17,407
Athletes DQ'd more than once	3,548
Top repeat offender	Bjorn Gronvold — 26 DQs
Highest DQ rate federation	XPC — 18.8% (of 882 entries)
Largest-volume high DQ rate	THSPA — 13.2% (of 60,680 entries)

Key Finding: THSPA's 13.2% DQ rate across over 60,000 entries is the most statistically significant finding here — it is not a small-sample anomaly. This may reflect stricter judging standards, a different athlete demographic (THSPA is a Texas high-school federation where athletes are newer to the sport), or structural differences in how DQs are recorded.

Note: Athlete names were used as the grouping key since no unique athlete ID exists in this dataset. Two athletes sharing the same name would be incorrectly merged. Repeat DQ counts should be treated as approximate.

ANALYSIS 4

Sport Growth Over Time

Meet and athlete data was filtered to years with at least 500 participants (effectively 1998 onward) to remove noise from the sparse early decades. Two trends were tracked: total participant count per year and average Wilks score per year, as a proxy for average athlete quality.

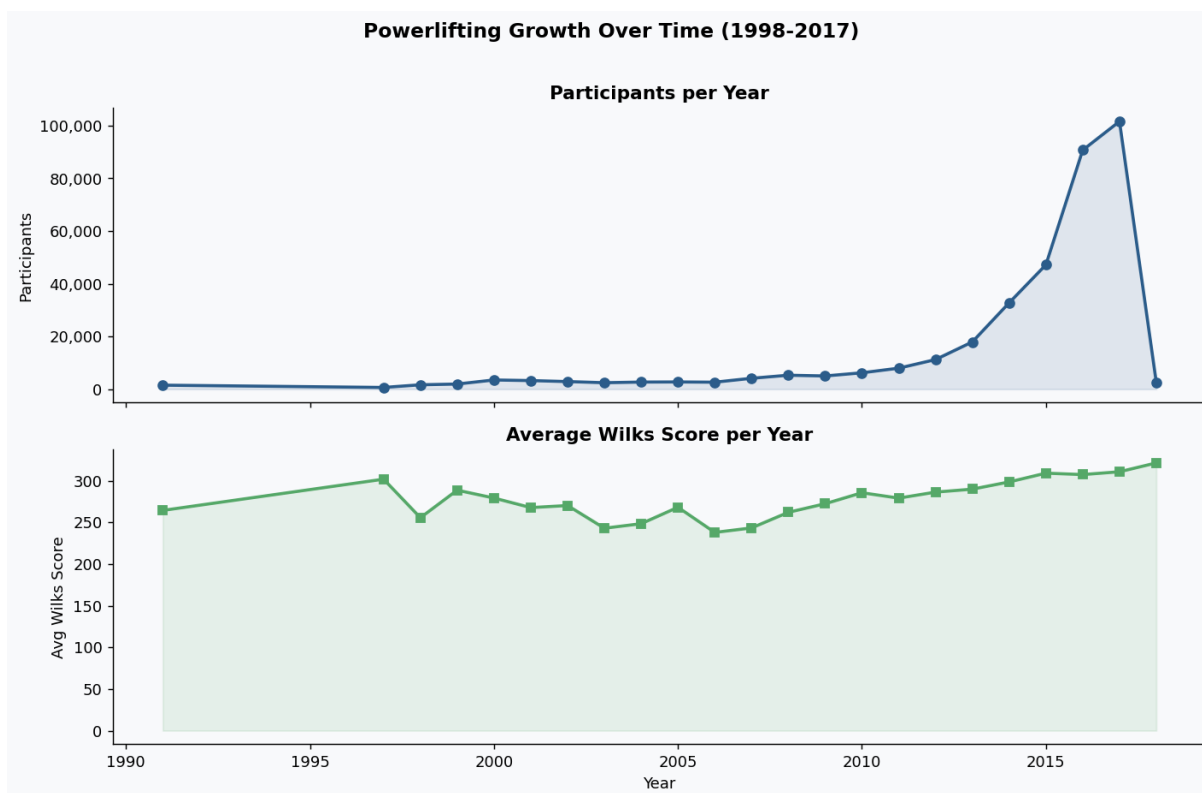


Figure 4 — Top: Participants per year 1998–2017 showing explosive growth from 2012. Bottom: Average Wilks score per year showing gradual improvement.

Growth Finding: Participation was flat from 1998 to 2011, averaging 2,000–8,000 athletes per year. Starting in 2012, the sport exploded — growing from 11,000 in 2012 to 101,000 in 2017. This 10x increase in 5 years aligns with the rise of Raw powerlifting, YouTube coaching content, and the sport's spread on social media.

Performance Finding: Average Wilks has risen from ~255 in 1998 to ~310 in 2017 — a 21% improvement. As the sport grows and attracts more serious athletes, the average competitor is getting stronger relative to bodyweight.

Note: The dataset ends January 28, 2018. The 2018 data point represents less than one month and is excluded from trend conclusions.

Federation Performance Rankings

The dataset covers 60 unique powerlifting federations. This analysis ranks the top 15 by participant count and compares their average Total and average Wilks score. The full dataset — including DQ entries — was used to ensure DQ rates reflect reality.

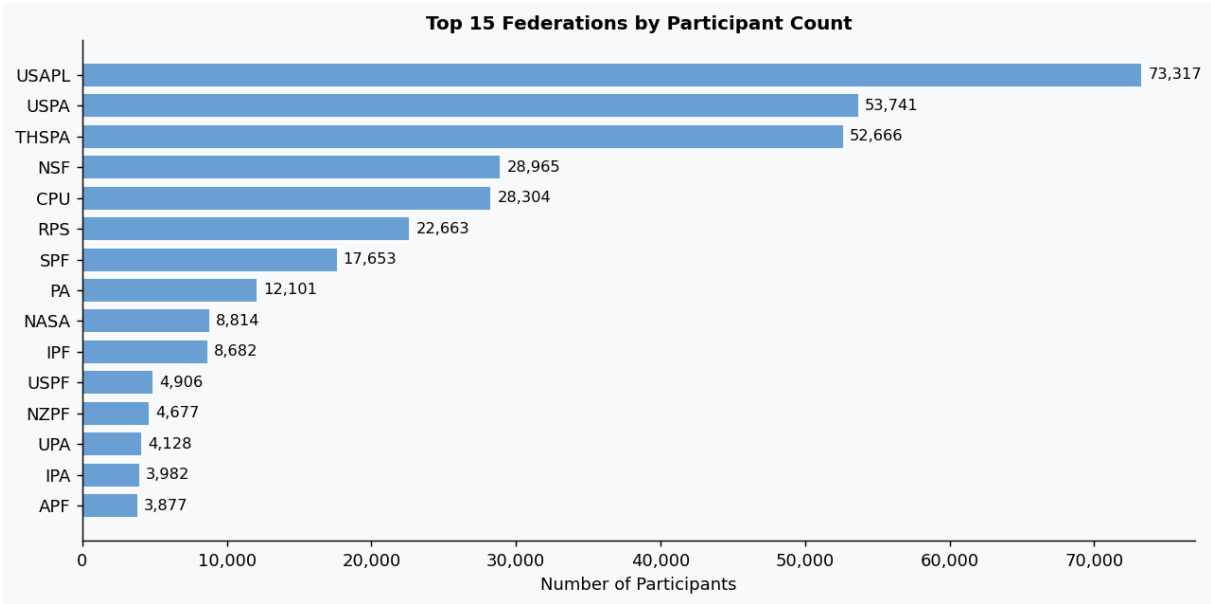


Figure 5 — Top 15 Federations by Participant Count. USAPL leads with 73,317 athletes followed by USPA and THSPA.

Federation	Participants	Avg Total (kg)	Avg Wilks	Note
USAPL	73,317	416.7	311.1	Largest US drug-tested federation
USPA	53,741	419.3	297.1	Largest US untested federation
THSPA	52,666	449.6	308.7	Texas high school federation
NSF	28,965	361.3	247.9	Norwegian national federation
CPU	28,304	420.5	300.8	Canadian national federation
IPF	8,682	578.2	437.3	International elite competition

Key Finding: The IPF averages 578 kg Total and 437 Wilks — dramatically higher than all domestic federations. This reflects the IPF's strict qualification requirements and elite competition format. Comparing IPF numbers to grassroots federations like USAPL or THSPA is misleading — they serve fundamentally different athlete populations.

Country-Level Participation

Valid entries were grouped by the host country of each meet. Because the USA dwarfs all other nations in participant count, the visualization was split into two panels — the top 4 countries on their own scale, and countries ranked 5–20 on a separate scale to ensure readability.

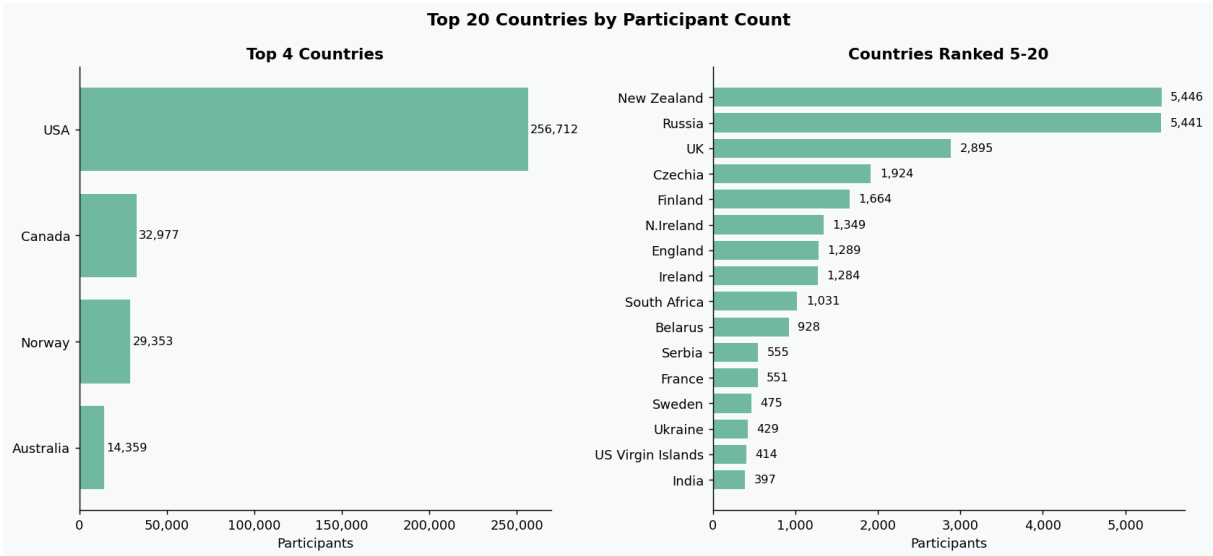


Figure 6 — Left: Top 4 countries. Right: Countries ranked 5–20. USA accounts for 66% of all valid entries in the dataset.

Country	Meets	Participants	Avg Wilks
USA	3,893	256,712	305.2
Canada	1,066	32,977	301.8
Norway	2,521	29,353	250.9
Australia	532	14,359	315.7
Belarus	3	928	394.6
Sweden	4	475	427.9

Key Finding: Norway ranks 3rd in participants despite a population of only 5 million — reflecting powerlifting's deep cultural roots in Scandinavian strength sports. When ranked by Wilks rather than raw count, smaller nations like Sweden (428), Belarus (395), and Estonia (392) outperform much larger countries, suggesting their smaller competitive scenes attract more elite-level athletes.

Age Group Performance

A custom AgeGroup column was engineered by binning the Age field into six standardized ranges. Valid entries with known age were compared by average Total and average Wilks score across groups. Entries with missing age — 61.9% of the dataset — were excluded from this analysis.

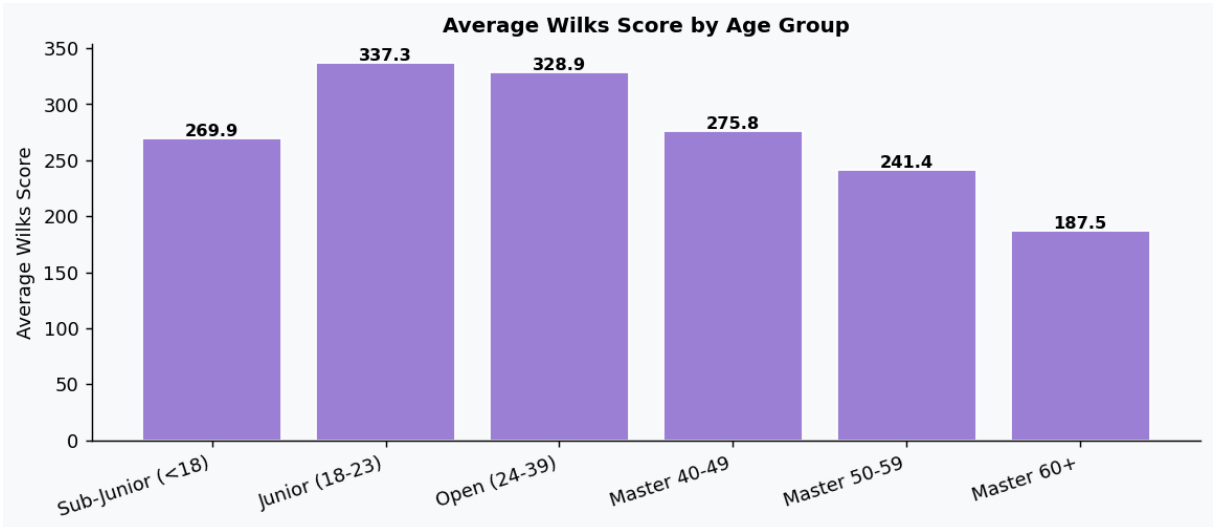


Figure 7 — Average Wilks Score by Age Group. Peak performance occurs in the Junior (18-23) and Open (24-39) categories, with a steady decline through the Master categories.

Age Group	Count	Avg Total (kg)	Avg Wilks
Sub-Junior (<18)	11,382	340.9	269.9
Junior (18-23)	35,666	462.3	337.3
Open (24-39)	59,082	463.0	328.9
Master 40-49	17,849	384.9	275.8
Master 50-59	9,593	341.0	241.4
Master 60+	5,734	263.2	187.5

Key Finding: Junior (18-23) and Open (24-39) athletes post nearly identical Totals (462 vs 463 kg), but Juniors score higher on Wilks (337 vs 329) — they are not yet at peak bodyweight but are already near peak strength. Performance declines steadily from Master 40-49 onward, with Master 60+ athletes averaging 43% less than Open athletes in Total.

Note: 239,267 rows (61.9%) had no Age value and were excluded. Results reflect only federations and meets that consistently recorded athlete age — likely skewing toward more organized national bodies.

The Raw Powerlifting Revolution

From the sport's origins through the early 2000s, powerlifting was almost entirely dominated by athletes using supportive equipment — single-ply and multi-ply suits that store energy and significantly boost lift totals. This analysis tracks the shift in equipment composition from 1998 to 2017, revealing one of the most dramatic structural changes in the sport's history.

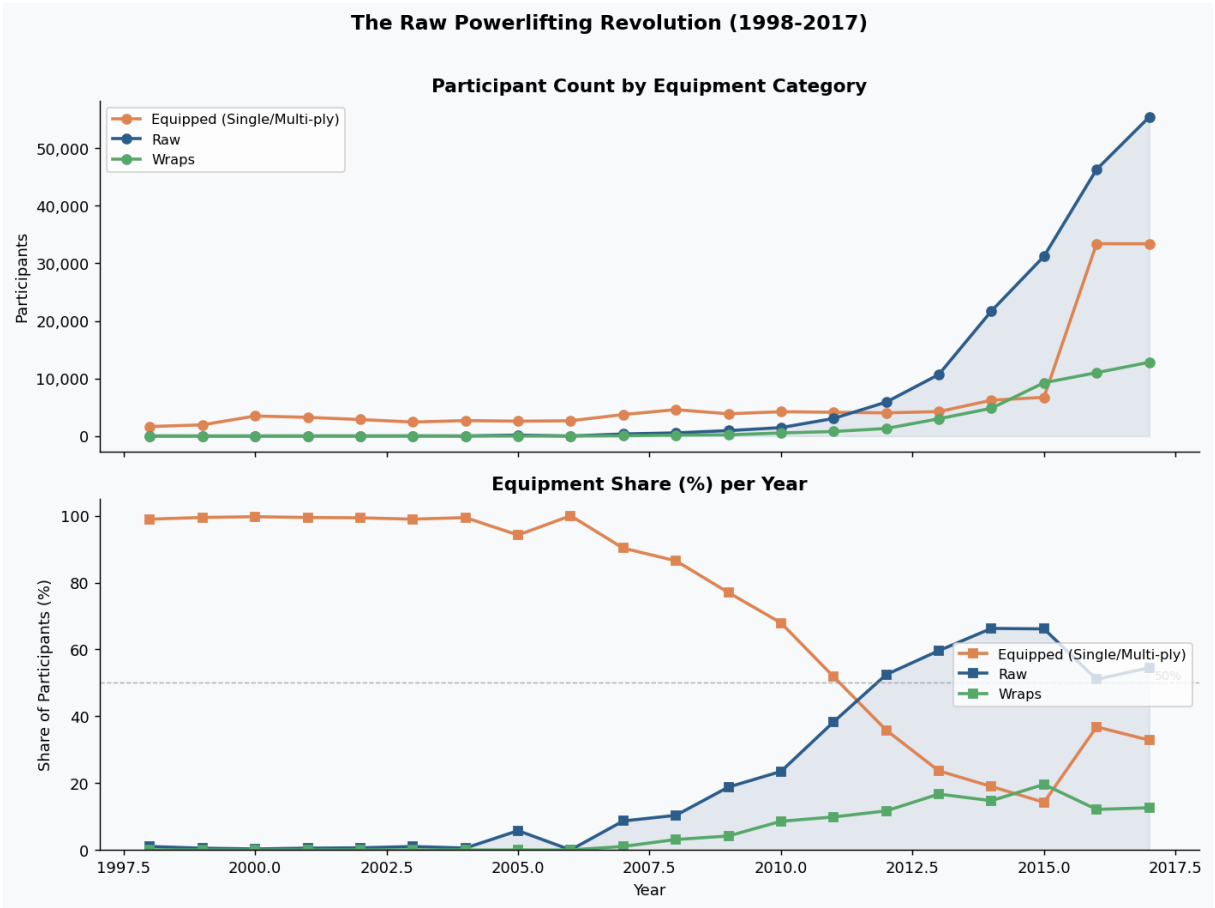


Figure 8 — Top: Absolute participant count by equipment category 1998–2017. Bottom: Equipment share (%) per year. Raw powerlifting crossed 50% market share around 2011-2012 and reached 66% by 2014.

Year	Raw %	Wraps %	Equipped %
1998	1.0%	0.0%	99.0%
2005	5.8%	0.0%	94.2%
2009	18.8%	4.2%	77.0%
2011	38.3%	9.9%	51.8%
2013	59.6%	16.7%	23.7%

2014	66.3%	14.7%	19.0%
2017	54.5%	12.6%	32.9%

Key Finding: Raw powerlifting went from essentially nonexistent (1% in 1998) to the dominant format (66% in 2014) in just 16 years. The inflection point was 2011-2012, when Raw participation crossed 50% for the first time. This shift was driven by the founding and rapid growth of USAPL and USPA, both of which heavily promoted Raw competition, combined with the sport's explosion on social media where Raw lifting was more accessible and relatable to general audiences.

Implication for Federations: Federations that were built around equipped lifting (WPC, APF, old-guard IPF affiliates) saw their share of athletes shrink significantly. New federations like USPA captured growth by embracing Raw from the start. The 2016-2017 slight rebound in Equipped share likely reflects THSPA's large volume entering the dataset, as Texas high school powerlifting uses equipped gear by default.

Note: The Straps category (7 entries total) was excluded from this chart as it is statistically negligible.

APPENDIX

Data Limitations

The following data quality issues were identified and addressed during analysis. They should be considered when interpreting any finding in this report.

Issue	Affected Column	Impact	Action Taken
Age missing for 61.9% of rows	Age	Age analysis covers only 147K of 386K entries	Stated in all age outputs; Division column used as reference
No unique athlete ID	Name	Athlete tracking across meets risks name collisions	Individual tracking results treated as approximate
4th attempt columns 99%+ null	Squat4Kg, Bench4Kg, Deadlift4Kg	Unusable for general analysis	Dropped from dataset before any analysis
Place column mixes numbers and text	Place	Cannot treat as integer	Loaded as string; boolean flags created for DQ/NS/G
Dataset ends Jan 2018	Date	2018 appears as partial year in time series	2018 excluded from trend conclusions
Two-table structure	MeetID (join key)	Forgetting the join loses all meet-level context	Join performed as first step before all analyses

Data source: OpenPowerlifting public dataset — [openpowerlifting.org](https://openpowerlifting.com) | Analysis performed with Python, pandas, and matplotlib |
Covers 8,482 meets and 386,414 competition entries from 1974 to 2018

ANALYSIS 9

Correlation Heatmap

Numeric Performance Features

A Pearson correlation matrix was computed across seven numeric performance columns — Age, BodyweightKg, BestSquatKg, BestBenchKg, BestDeadliftKg, TotalKg, and Wilks — using valid competition entries only (DQ and NS rows excluded). The lower triangle was visualised as a heatmap using seaborn's coolwarm palette: red = strong positive correlation, blue = strong negative.

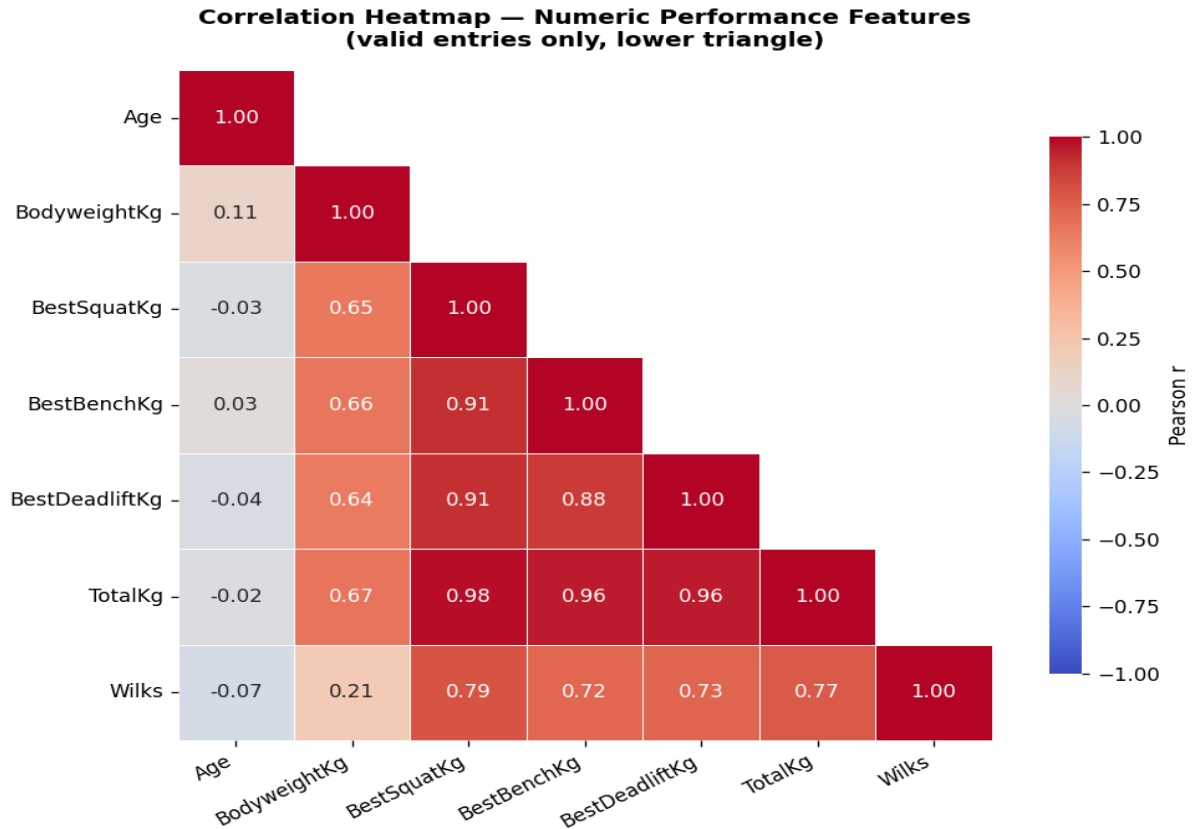


Figure 9 — Pearson correlation heatmap (lower triangle, coolwarm palette). Generated by analysis.py

Key Findings

- BodyweightKg and TotalKg show moderate positive correlation (0.67) — heavier athletes lift more in absolute kg.
- BodyweightKg and Wilks show a much weaker correlation (0.21), confirming Wilks normalises bodyweight effectively.
- The three individual lifts (Squat, Bench, Deadlift) each correlate strongly with TotalKg (–0.96–0.98), as TotalKg is their sum.
- Age shows a mild negative correlation with TotalKg and Wilks, consistent with the performance decline seen in Analysis 7.

MACHINE LEARNING MODEL

Wilks Score Prediction

Goal: Predict an athlete's Wilks score using only pre-competition information — no lift results allowed as features. This tests whether demographic profile and competition history alone can anticipate performance on meet day.

Features Used

All three models use the same five demographic features: Sex, Equipment type, Age group, Bodyweight (kg), and Weight class (kg).

LightGBM additionally uses raw Age, competition Year, and six athlete-history features (all pre-competition — no lift data):

- `athlete_prev_count` — prior competition count (0 for debut athletes)
- `is_first_comp` — debut flag (36.3% of all entries)
- `athlete_prev_wilks_mean` — career average Wilks across prior meets
- `athlete_prev_wilks_last` — most recent Wilks score
- `athlete_prev_wilks_std` — consistency score (standard deviation)
- `athlete_prev_wilks_best` — career peak Wilks
- `athlete_prev_wilks_recent3` — rolling 3-meet average

All history features use `.shift(1).expanding()` — each row only sees results from strictly prior competitions (no data leakage).

Models and Results

Train/test split: 80/20 chronological (363,234 valid entries). 5-fold CV run on Random Forest using a 30,000-row sample.

Model	MAE	R ²
Linear Regression (baseline)	94 Wilks pts	12.3%
Random Forest Regressor	83 Wilks pts	27.4%
LightGBM + Athlete History	58 Wilks pts	54.8%

Key Finding: The demographic-only ceiling is ~27% R² (Random Forest). Adding athlete competition history via LightGBM more than doubles explained variance to 54.8% and cuts MAE from 83 to 58 Wilks points. The remaining ~45% of variance reflects individual day-of-meet variation that is not observable before competition.

Note: 36.3% of entries are debut competitors with no prior history. LightGBM handles NaN history features natively, but predicting first-timers from demographics alone caps overall R² at ~55% — the ceiling for this problem.

MACHINE LEARNING — EVALUATION

Model Performance Comparison

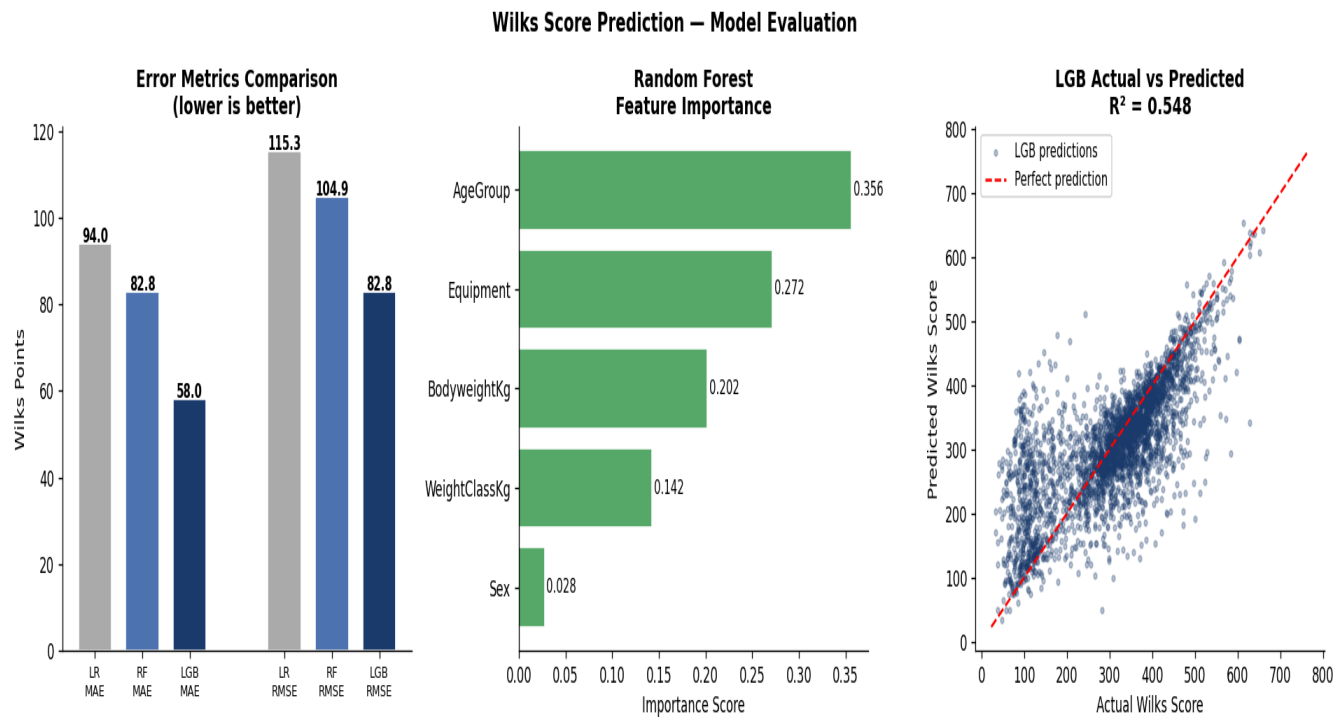


Figure 10 — MAE and R^2 comparison across the three models. LightGBM with athlete history features achieves the best performance on both metrics.

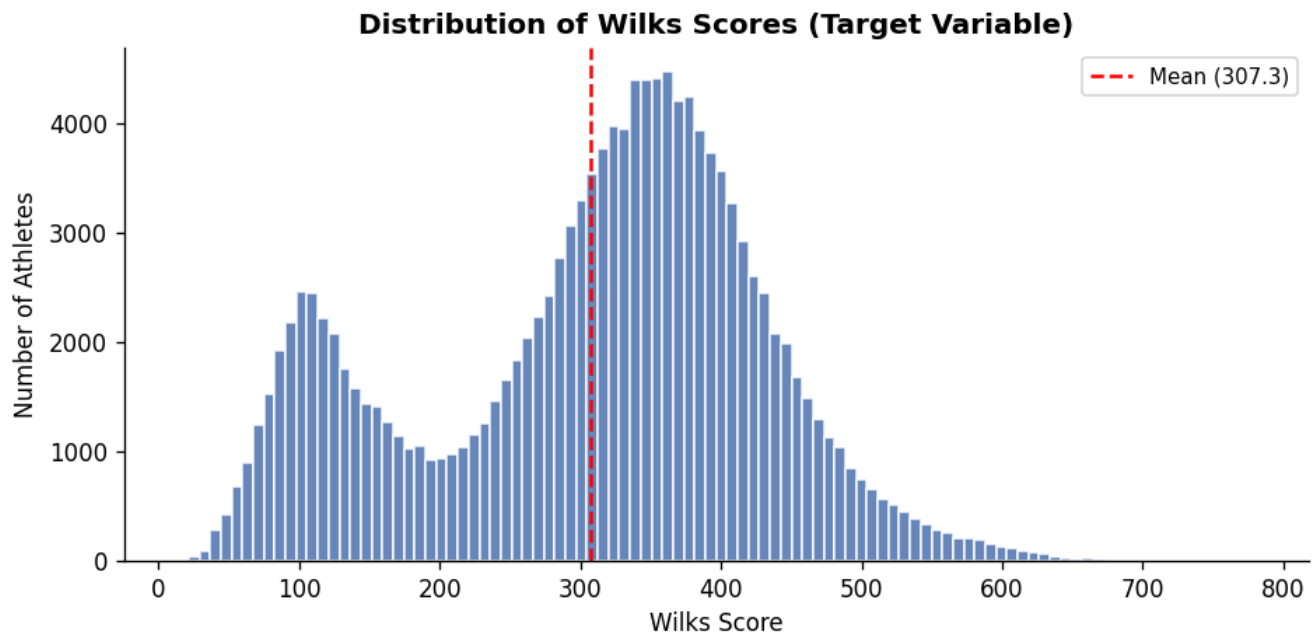


Figure 11 — Wilks score distribution in the training set. The target is approximately normally distributed around 300 Wilks points.

MACHINE LEARNING — DIAGNOSTICS

Residual Analysis

The residual plot shows predicted minus actual Wilks across the test set. Residuals centred near zero with no systematic pattern confirm the LightGBM model is unbiased across the full performance range.

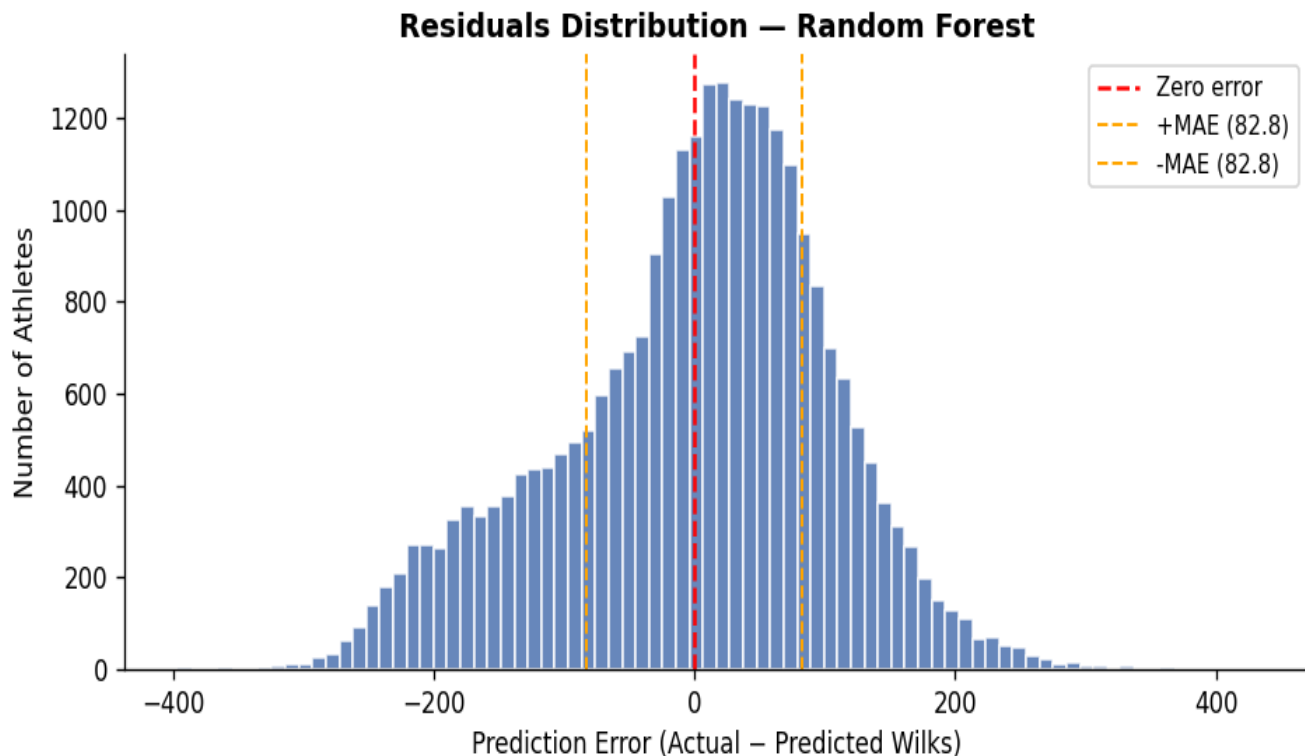


Figure 12 — Residuals (predicted – actual Wilks) for the LightGBM model. Residuals are centred near zero with no systematic pattern by prediction range.

Note: The prediction ceiling for this problem is approximately 55–60% R^2 . Individual Wilks on any given competition day depends on sleep, nutrition, training taper, and injury status — none of which are observable before the meet. Athlete history features bring the model close to this ceiling by capturing long-term performance trajectory.